

WHEELING OHIO COUNTY AP, WV, USA

WMO: 724275

Lat: 40.176N

Lon: 80.647W

Elev: 359

StdP: 97.08

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 14894

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.9	-12.3	-19.9	0.7	-13.1	-17.7	0.8	-11.0	11.4	3.5	10.4	2.0	3.6	260	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	31.2	22.5	29.8	21.8	28.6	21.2	24.0	28.8	23.2	27.7	22.5	26.7	3.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.9	26.6	21.9	17.3	25.8	21.2	16.5	25.0	73.6	29.0	70.5	27.9	67.7	26.6	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.5	-17.6	32.7	2.9	1.7	-19.7	34.0	-21.5	35.0	-23.1	36.0	-25.2	37.3		
(5)			-18.1	25.4	2.9	0.9	-20.1	26.1	-21.8	26.6	-23.4	27.1	-25.5	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-1.5	0.1	4.7	11.5	16.7	20.5	22.7	22.1	18.8	12.5	6.6	1.4
	DBStd	9.90	6.90	6.36	6.32	5.33	4.46	3.32	2.75	2.70	3.86	5.02	5.44	5.94
	HDD10.0	1319	362	283	189	44	4	0	0	0	0	31	131	274
	HDD18.3	2978	616	510	422	211	86	17	2	3	40	192	353	526
	CDD10.0	1830	5	6	26	91	211	316	393	376	264	108	28	7
	CDD18.3	448	0	0	1	7	35	83	137	121	54	10	1	0
	CDH23.3	3226	0	1	5	61	271	577	1017	846	389	57	2	0
	CDH26.7	832	0	0	0	7	48	138	301	230	99	9	0	0
Wind	WSAvg	3.4	4.3	4.2	4.1	4.1	3.2	2.9	2.6	2.4	2.7	3.3	3.8	4.0
Precipitation	PrecAvg	1014	82	63	88	90	106	103	104	89	82	67	68	73
	PrecMax	1364	180	138	147	120	158	245	205	163	235	145	106	163
	PrecMin	793	37	32	33	40	34	31	65	41	29	17	15	34
	PrecStd	141	39	26	32	23	37	44	36	39	46	32	24	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	19.5	23.5	27.6	29.8	31.4	32.9	32.6	31.5	28.0	22.1	18.0
		MCWB	13.0	12.9	14.7	16.9	20.6	22.0	23.8	22.5	21.6	20.3	15.4	13.9
	2%	DB	14.6	16.0	20.8	25.2	27.9	29.6	31.2	30.9	29.3	24.7	19.4	15.5
		MCWB	11.6	10.9	13.0	15.3	19.5	21.6	22.9	22.4	21.0	17.9	13.5	12.1
	5%	DB	11.9	12.9	17.8	22.8	26.4	28.3	29.7	29.1	27.5	22.5	17.3	13.0
		MCWB	9.6	9.3	11.4	14.5	18.9	21.2	22.3	21.5	20.4	16.6	12.3	10.0
	10%	DB	8.4	9.9	14.9	20.6	24.6	27.0	28.2	27.6	25.6	20.6	15.1	10.5
		MCWB	5.9	6.1	10.1	13.1	17.9	20.6	21.5	20.9	19.3	16.0	11.1	7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	14.4	15.9	18.5	22.5	24.0	25.4	24.8	23.9	21.5	16.9	15.0
		MCDB	16.7	18.5	21.7	24.8	28.1	28.7	30.7	30.1	29.0	26.3	20.3	17.4
	2%	WB	12.2	12.1	14.2	16.9	20.9	23.0	24.2	23.7	22.5	19.7	14.9	12.8
		MCDB	14.3	14.8	19.0	22.8	26.1	27.8	29.1	28.3	27.0	22.9	18.2	14.8
	5%	WB	9.7	9.5	12.7	15.7	19.9	22.1	23.4	22.9	21.4	18.0	13.2	10.7
		MCDB	11.5	13.2	16.7	20.6	24.7	26.9	27.9	26.9	25.2	21.1	16.2	12.7
	10%	WB	6.1	6.5	10.5	14.5	18.9	21.4	22.6	22.1	20.5	16.3	11.5	7.8
		MCDB	8.3	9.7	14.4	19.0	23.3	25.6	26.7	25.8	23.9	19.9	14.7	10.0
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.3	9.6	10.9	10.5	9.8	9.8	10.0	10.4	9.5	8.5	7.1
		MCDBR	10.1	12.0	13.0	13.2	12.2	11.1	11.1	11.5	12.0	11.6	11.2	10.1
		MCWBR	8.3	9.2	7.8	7.2	6.3	5.4	4.6	4.7	5.4	6.3	7.5	8.3
	5% WB	MCDBR	9.7	11.7	11.3	11.2	10.4	9.5	9.6	9.2	9.7	9.9	9.7	10.0
		MCWBR	8.7	9.9	7.8	7.2	6.0	5.3	4.6	4.5	4.9	6.1	7.7	9.1
Clear-Sky Solar Irradiance	taub	0.301	0.322	0.347	0.381	0.423	0.452	0.462	0.460	0.394	0.361	0.318	0.299	
	taud	2.473	2.393	2.397	2.330	2.221	2.173	2.167	2.153	2.317	2.413	2.512	2.518	
	Ebn at Noon	867	894	906	893	858	830	818	807	843	836	840	844	
	Edh at Noon	77	96	107	123	140	147	147	145	115	94	74	69	
All-Sky Solar Radiation	RadAvg	1.55	2.40	3.39	4.56	5.32	5.79	5.74	5.27	4.31	2.91	1.97	1.33	
	RadStd	0.15	0.35	0.31	0.44	0.44	0.45	0.39	0.30	0.42	0.29	0.27	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+125	+62

Nomenclature: See separate page